JUNIOR CANADIAN CHEMISTRY OLYMPIAD

British Columbia Chemistry Olympiad

2023

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.
Write your answers on the back side of the Answer Sheet provided. Do not forget to write your name and your school name on the front of the Answer Sheet. While students are expected to attempt all questions for a complete examination in 1.5 hours, it is recognized that backgrounds will vary and students will not be eliminated from further competitions because they have missed parts of the paper.

1) The density of carbon tetrachloride at 25 °C is 1.59 g/cm^3. What is the volume of 1 mole of CCl_4 at 25 °C?

a) 75.1 cm^3 b) 9.67 cm^3 c) 89.2 cm^3 d) 1.03 cm^3 e) 96.7 cm^3

2) Eugenol ($C_{10}H_{12}O_2$) has been shown to have antibacterial, antifungal, antioxidant and antineoplastic activity. Clove oils including eugenol have been claimed to have gentle local anesthetic and antiseptic activities and previously were commonly used in dentistry. Eugenol is the major constituent (80%) in the aromatic oil extract from cloves. How many moles of eugenol are there in 1 kg of clove oil extract?

a) 0.257 b) 4.87 c) 6.09 d) 7.61 e) 13.1

3) What is the type of the following reaction?

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

a) Synthesis
b) Single replacement
c) Decomposition
d) Neutralization
e) Combustion

4) After balancing the following reaction, the stoichiometric coefficient of MnO_4^- is 2 and the stoichiometric coefficient of NO_2^- is 5. What is the stoichiometric coefficient of H^+?

$$NO_2^- + MnO_4^- + H^+ \rightarrow NO_3^- + Mn^{2+} + H_2O$$

a) 2 b) 3 c) 4 d) 5 e) 6

5) Which of the following chemical equations is properly balanced?

a) $C_6H_{12}O_6 + 12O_2 \rightarrow 6H_2O + 6CO_2$
b) $C_6H_8 + 6O_2 \rightarrow 5CO_2 + 4H_2O$
c) $4NH_2CH_2COOH + 9O_2 \rightarrow 8CO_2 + 10H_2O + 2N_2$
d) $C_2H_2 + 2O_2 \rightarrow 2CO_2 + H_2O$
e) $C_{12}H_{22}O_{12} + 2H_2O \rightarrow 2C_6H_{12}O_6$

6) Menthol has local anesthetic and counterirritant qualities, and it is widely used to relieve minor throat irritation. Yearly world production of menthol is about 12,000 tons. Due to its volatility, the qualitative experience of menthol may be modulated by its preparation and combination with other compounds. One such method of preparation is dilution. How many grams of menthol (156.26 g/mol) is needed to prepare a 0.1M solution in ethanol in a volumetric flask of 200 mL?

a) 12.8 b) 7.81 c) 3.13 d) 7.61 e) 13.1

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7) German chemist Eilhard Mitscherlich discovered in 1844 that solutions of salts of tartaric acid affected polarised light in a surprising way. Residues deposited during wine-making always rotated the plane polarized light to the right while tartaric acid synthesised in the laboratory had no effect on polarized light. Yet the two acids where chemically indistinguishable. Louis Pasteur, born 200 years ago, noticed something Mitscherlich had missed. Pasteur spotted that one salt, sodium ammonium racemate, crystallises in two geometrical forms which are mirror-images of each other. When he separated them (painstakingly picking out individual crystals with tweezers), and then dissolved them in two separate solutions in water, both solutions rotated the plane of polarised light, in opposite directions! Today, this phenomenon is called optical activity and measuring the specific rotation of optically active compounds is important for organic chemistry. The specific rotation of a compound is calculated as follows:

$$[\alpha] = \frac{\alpha}{l[X]}$$

Where $[\alpha]$ is the specific rotation of the compound, α is the rotation of the plane polarized light in degrees, l the path length of the cell in dm and [X] is the concentration of the compound in g/mL. 2.89 g of one of the optically active crystals separated by Pasteur is dissolved in 10 mL of water and placed in a cell of 1 dm, the observed rotation of polarized light is −10°. What is the specific rotation of the crystal?

a) −34.6° b) +28.9° c) −28.9° d) −10° e) +34.6°

8) Three acids that can be found in a household are acetic acid (AA), citric acid (CA), and bisulfate (BSA). The pK_a values of the three acids are 4.76, 2.79, and 1.92, respectively. Which option correctly ranks these acids in the order of ***decreasing*** acid strength?

a) AA > CA > BSA
b) CA > BSA > AA
c) AA > BSA > CA
d) BSA > CA > AA
e) BSA > AA > CA

9) Ultraviolet/Visible (UV/Vis) spectroscopy is routinely used in analytical chemistry for the quantitative determination of diverse analytes. In a typical UV/Vis experiment, attenuation is measured for light traveling through a sample solution. The light attenuation can be used to calculate the absorbance of the solution. The Beer–Lambert law relates the absorbance to the molar absorptivity of the molecule, the path length of the light and the concentration of the solution:

$$A = \varepsilon l c$$

Where A is absorbance, ε is the molar absorptivity (a property of the molecule), and l is the path length of the light through the sample and c the concentration of the solution.
What is the concentration for a Blue #1 dye solution with an absorbance of 0.49 and a molar absorptivity coefficient of 130,000 $M^{-1}cm^{-1}$, measured in a cuvette with a path length of 0.25 cm at 630 nm?

a) 15 µM b) 3.8 nM c) 15 mM d) 0.94 µM e) 3.8 µM

10) Airbags contain a mixture of sodium azide, potassium nitrate, and silicon dioxide. A sensor detects a head on collision that cause the sodium azide to be ignited and to decompose forming sodium oxide and nitrogen gas as follows:

$$10NaN_3 + 2KNO_3 \rightarrow 5Na_2O(s) + K_2O(s) + 16N_2(g)$$

The average volume of an airbag is 140 L. Assuming that the airbag is filled at 1 atm at a temperature of 298 K, how many moles of NaN_3 are needed on average? (assume ideal gas behavior)

a) 5.73 b) 3.58 c) 3.91 d) 17.7 e) 25.9

11) Which of the following atoms has the largest first ionization energy?

a) Si b) F c) K d) Be e) Cl

12) Which of the following has the greatest bond order?

a) O_2^{2+} b) O_2^{+} c) O_2 d) O_2^{-} e) O_2^{2-}

13) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value. A good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{eff} = Z – S$, where Z is the atomic number of the atom and S a screening constant. In the second row of the periodic table, the screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons, excluding the electron for which the effective nuclear charge is to be determined. Use Slater's rule to estimate the Z_{eff} for a valence electron in neon (Ne).

a) 3.35 b) 5.85 c) 2.35 d) 6.85 e) 4.15

14) While preparing a solution of KNO_3 in water, a chemistry student noticed that the volumetric flask felt cold. Intrigued, the student decided to perform a rudimentary experiment to measure the enthalpy of solution (ΔH) of KNO_3. For his experiment, the student added 10.1 g KNO_3 to 180 g of water and used a thermometer to record the temperature. A temperature change of −4.6 degrees was recorded. Assuming that the heat capacity of water is unchanged at 75 $JK^{-1}mol^{-1}$, what is the enthalpy of solution for KNO_3?

a) −34.5 kJ/mol
b) 62.1 kJ/mol
c) 34.5 kJ/mol
d) −6.15 kJ/mol
e) −62.1 kJ/mol

15) An element X forms compounds with chlorine as dichloride (XCl_2), which is a white crystalline solid, and tetrachloride (XCl_4), a colorless liquid. Treatment of 10.00 g XCl_2 with excess chlorine forms 13.74 g XCl_4. Identify X.

a) Pt b) Ti c) S d) Sn e) Rh

16) Following the VSEPR theory, which of the following molecules has a molecular geometry that is trigonal pyramidal?

a) $[BrO_4]^-$
b) $[BrO_3]^-$
c) $[BrO_2]^-$
d) BrO_2
e) Br_2O

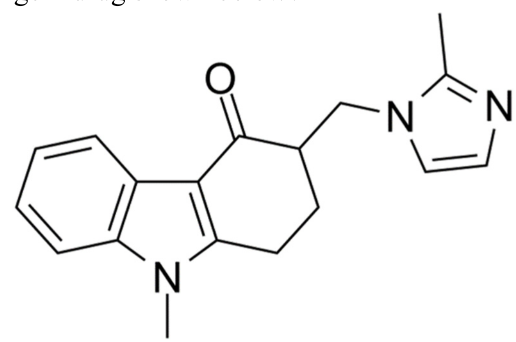
17) Which species is the reducing agent in the following reaction?

$$4H^+(aq) + 2Cl^-(aq) + MnO_2(s) \rightarrow Cl_2(g) + Mn^{2+}(aq) + 2H_2O(l)$$

a) H^+ b) Mn^{2+} c) MnO_2 d) Cl_2 e) Cl^-

18) The drug shown below is a medication used to prevent nausea and vomiting caused by cancer chemotherapy, radiation therapy, or surgery. Which of the following molecular formula is the correct one for the antiemetic agent drug shown below?

a) $C_{18}H_{18}N_3O$
b) $C_{17}H_{16}N_3O$
c) $C_{16}H_{13}N_3O$
d) $C_{19}H_{18}N_3O$
e) $C_{18}H_{19}N_3O$



19) Which of the following is the correct order of ***increasing*** melting point?

a) NaCl(s), $C_{16}H_{34}$ (s), MgO(s), C(s), Al(s)
b) $C_{16}H_{34}$(s), Al(s), NaCl(s), MgO(s), C(s)
c) $C_{16}H_{34}$ (s), C(s), MgO(s), NaCl(s), Al(s)
d) C(s), $C_{16}H_{34}$ (s), NaCl(s), Al(s), MgO(s)
e) $C_{16}H_{34}$ (s), MgO(s), C(s), Al(s), NaCl(s)

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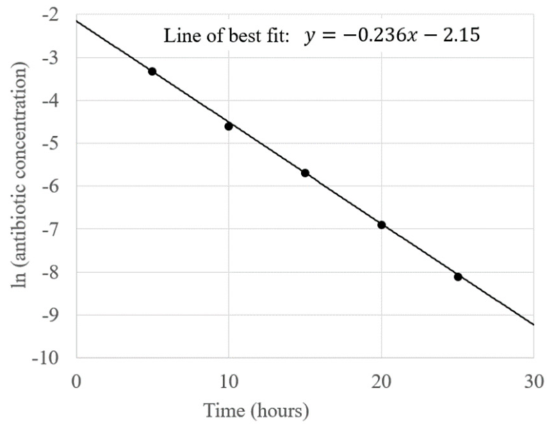
20) The following plot records the process of the elimination of an antibiotic by the human body. What is the half life of the antibiotic?



a) 4.24 h
b) 0.24 h
c) 2.94 h
d) 0.34 h
e) 2.15 h

21) What is the pH of a 0.350 M solution of sodium citrate at 25 °C (pKb of citrate = 7.60)?

a) 4.03 b) 9.97 c) 8.45 d) 5.19 e) 3.28

22) The atomic radius of Zirconium (Zr) is 155 pm. The atomic radius of Hafnium (Hf) is also 155 pm. Which of the following is the best explanation of this phenomenon? (*Hint: look at the periodic table and write down the electron configuration. Slater's rule does not apply to atoms lower than the fourth row*)

a) The ionization energy of Hf is lower than the ionization energy of Zr.
b) Hafnium has more protons, therefore the attraction to the valence electrons from the nucleus is lower.
c) Hafnium has a greater effective nuclear charge and therefore the valence electrons experience stronger attraction from the nucleus.
d) Zirconium has a greater effective nuclear charge and therefore the valence electrons experience stronger attraction from the nucleus.
e) The highest sub-energy level in an atom of Zirconium is 5s while in an atom of Hafnium is 6s.

23) A liquid organic compound X has a molar mass of less than 150 g/mol. A sample of 11.8 g of this compound completely combusted in 20.0 L of oxygen to produce carbon dioxide and water vapor. After the reaction, the gas phase of the reaction mixture was passed through a concentrated $H_2SO_4(aq)$ solution ($H_2SO_4(aq)$ is a good dehydrating agent). The remaining gas had a volume of 17.76 L. This gas was then passed through solid CaO, which reduced the volume of the gas by 11.20 L. Identify X. (assume all reactions were complete and all gas volumes are based on a pressure of 1 atm and a temperature of 273.15K).

a) $C_6H_{12}O_3$ b) $C_6H_{10}O_4$ c) $C_5H_{10}O_3$ d) $C_6H_{14}O_2$ e) $C_7H_{12}O_2$

24) A gas contains hydrogen and other molecules under an atmospheric pressure of 745 mmHg and a temperature of 120.2 ºC. what can be said about the kinetic energy?

a) All molecules in the gas will have a lower kinetic energy than hydrogen molecules.
b) All molecules in the gas will have a kinetic energy of zero.
c) All molecules in the gas will have a different kinetic energy due to their different molar mass.
d) All molecules in the gas will have exactly the same kinetic energy.
e) All molecules in the gas will have the same average kinetic energy.

25) Esterification is an important organic reaction. Using acid as a catalyst, which of the following will result in an esterification reaction to produce ethyl acetate.

a) $CH_3CH_2OH + CH_3COCH_3$
b) $CH_3CHO + CH_3COOH$
c) $CH_3CH_2OH + CH_3OCH_3$
d) $CH_3COOH + CH_3OCH_3$
e) $CH_3COOH + CH_3CH_2OH$

End of examination questions.